CV AKADEMIK

Hafidh Khoerul Fata

Dosen Matematika · Universitas Diponegoro · Sistem dinamik nonlinear dan bifurkasi.

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Profil profesional

PERAN

Dosen Matematika

AFILIASI

Universitas Diponegoro

JABATAN AKADEMIK

Asisten Ahli

GELAR PROFESI

Gr. · Matematika

PENDIDIKAN

2019–2021
Magister Matematika
Institut Teknologi Bandung

2018–2019
Pendidikan Profesi Guru · Matematika
Universitas Negeri Semarang · Gr.

2013–2017
Sarjana Matematika
Universitas Diponegoro

MINAT RISET

Sistem dinamik nonlinear, bifurkasi, efek delay dan memori, pemodelan matematika, dan dinamika numerik.

MSC2020

- **37-XX** Dynamical systems
- **37Gxx** Bifurcation theory
- **39Axx** Difference equations
- **34Kxx** Functional-differential equations
- **37Mxx** Numerical treatment

PROFIL AKADEMIK

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PROYEK TERPILIH

2024–2026

2026 · **KETUA**
Stability Analysis and Regime Change in Discrete Cyber Defense Model with Adaptive Patching

2026 · **ANGGOTA**
Nonlinear Dynamics and Chaos in a Banking Duopoly with Benchmark Deposit Rate Policy

2025 · **KETUA**
Exploration of Nonlinear Dynamical Systems: Analytical and Numerical Approaches to Unveil Complex Bifurcation Phenomena

2024 · **KETUA**
Analisis Bifurkasi Sistem Dinamik Non-Linier pada Model J.C. Sprott NE-14

Publikasi terpilih

14 JOURNAL ARTICLES · 1 PREPRINT · 1 BOOK

2026 Bifurcation structure and multistability in a sinusoidal modification of the Hénon map
Journal of Difference Equations and Applications · DOI 10.1080/10236198.2026.2689969

2026 Nonlinear Dynamics and Bifurcation Structure of a Stage-Structured Theta-Ricker Map with Cross-Stage Feedback
WSEAS Transactions on Mathematics · DOI 10.37394/23206.2026.25.3

2026 Deterministic and Stochastic Modeling of Deposit–Loan Dynamics with Optimal Regulatory Control
International Journal of Financial Studies · DOI 10.3390/ijfs14070174

2026 Threshold-Induced Bifurcation and Chaos in a One-Dimensional Coastal Risk Adaptation Map
World Journal of Advanced Research and Reviews · DOI 10.30574/wjarr.2026.30.3.1687

2025 Logistic Map-Based Banking Loan Dynamics with Central Bank Policies
Journal of Applied Nonlinear Dynamics · DOI 10.5890/JAND.2025.09.006

2025 A Study of Discrete-time Dynamical Systems with One Stable Equilibrium Using Standard and Nonstandard Difference Schemes
WSEAS Transactions on Mathematics · DOI 10.37394/23206.2025.24.64

Daftar publikasi lengkap ↗

PENGAJARAN

PROGRAM SARJANA

Analisis Riil
Limit, kontinuitas, barisan dan deret fungsi, serta integral.

Persamaan Diferensial Parsial
Klasifikasi, difusi, gelombang, dan metode penyelesaian analitik.

Algoritma & Pemrograman
Dekomposisi masalah, pseudocode, flowchart, dan dasar Python.

Matematika Teknik
Aljabar linear dan metode matematika untuk masalah teknik.

KREDENSIAL

AWS Academy Graduate – Generative AI Foundations

Badge pelatihan

Fakultas Bersertifikasi Gemini · Google for Education

MCE · Microsoft Certified Educator

PENGABDIAN

8 ACTIVITIES · 2023–2025

Delapan kegiatan pengabdian selesai, tujuh dipimpin sebagai ketua, meliputi pendampingan ilmiah, persiapan olimpiade matematika, visualisasi data, dan pengembangan potensi lokal.

- MYRES research mentoring for MAN Kendal students · 2025
- Gunung Pegat tourism revitalization masterplan · 2024
- Mathematics olympiad problem-solving preparation · 2025
- Stunting-data visualization and demographic mapping · 2023

Buku & kekayaan intelektual

2023–2025

BOOK · 2025

Teori Himpunan dan Relasi

UNDIP Press · ISBN 9786324176148 · 116 pages

IP · 2023

Banking Loan Dynamics with Dividend Payments

IP · 2023

Program Diagram Eksponen Lyapunov Dinamika Kredit Berdasarkan Perubahan Deviden

IP · 2023

Program Diagram Bifurkasi Dinamika Kredit Berdasarkan Perubahan Deviden